Cookie Puzzle

Aunt Dorothy knows that one of her, Bruno's, Carla's, Dorota's or Eduardo's stole all the cookies. She also knows that the guilty always lie and the innocent always tell the truth.

- Bruno says: "The culprit is either Eduardo or Dorota".
- Eduardo says: "The guilty person is a girl".
- Finally, Dorota says: "If Bruno is guilty, then Carla is innocent."

Who ate the cookies? Who is innocent?

Predicate Logic Solution

To solve this puzzle we use the Gamma-ND extension to the plugin **prettyprint**. The notation follows all necessary steps, starting from defining the objects and formalizing the statements in logic format, and continuing with the transformation of the Knowledge base and the application of the **resolution method** to derive the final conclusion.

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1. Vocabulary

Based on the problem statement, the main elements of the formal language can be identified: constants, variables, and predicates.

Constants

Constants represent the specific individuals mentioned in the problem.

Object	Constant	Information
Ann	a	female
Bruno	b	male
Carla	c	female
Dorota	d	female
Eduardo	e	male

Variables

- X, Y : a variable representing an arbitrary individual in the problem domain.
- Y : another variable representing an arbitrary individual in the problem domain.

Predicates

- $\text{guilty}(X)$: a predicate that is true if and only if the individual X is guilty.

2. Knowledge Base Formalization

At least one person is guilty
Formula

$$\exists X.\,\text{guilty}(X)$$

Exactly one person is guilty

$$\exists X.\Big(\text{guilty}(X) \wedge \forall Y.\big(Y = X\big) \rightarrow \text{guilty}(Y)\Big)$$

Bruno says: "The culprit is either Eduardo or Dorota".

$$\text{guilty}(e) \vee (\text{guilty}(e) \rightarrow \text{guilty}(d))$$

Eduardo says: "The guilty person is a girl".

$$\text{guilty}(e) \rightarrow (\text{female}(e) \wedge \text{guilty}(e))$$

Dorota says: "If Bruno is guilty, then Carla is innocent".

$$\text{guilty}(b) \rightarrow (\text{guilty}(b) \rightarrow \neg\text{guilty}(c))$$

3. CNF Transformation

At least one person is guilty
Formula

$$\exists X.\,\text{guilty}(X)$$

Conversion to CNF:

$$\exists X.\,\text{guilty}(X) \rightarrow \text{guilty}(a) \vee \text{guilty}(b) \vee \text{guilty}(c) \vee \text{guilty}(d) \vee \text{guilty}(e)$$

Resulting clauses:

$$\text{guilty}(a) \vee \text{guilty}(b) \vee \text{guilty}(c) \vee \text{guilty}(d) \vee \text{guilty}(e)$$

Exactly one person is guilty:
Formula

$$\exists X.\Big(\text{guilty}(X) \wedge \forall Y.\big(Y = X\big) \rightarrow \text{guilty}(Y)\Big)$$

Conversion to CNF:

- Introduce fresh quantifiers:
  - $X = Y \vee \neg(X = Y)$
  - $\exists X.\,\text{guilty}(X) \rightarrow \text{guilty}(a) \vee \text{guilty}(b) \vee \text{guilty}(c) \vee \text{guilty}(d) \vee \text{guilty}(e)$ (at least one guilty)
  - $\forall X.\,\forall Y.\,(X = Y) \vee \neg(X = Y) \rightarrow \text{guilty}(X) \wedge \neg\text{guilty}(Y) \rightarrow \text{guilty}(Y) \wedge \neg\text{guilty}(X)$ (at most one guilty)
- Combine them:

$$\begin{aligned}
&(\text{guilty}(a) \vee \text{guilty}(b) \vee \text{guilty}(c) \vee \text{guilty}(d) \vee \text{guilty}(e)) \\
&\wedge (\text{guilty}(a) \vee \neg\text{guilty}(b)) \\
&\wedge (\text{guilty}(a) \vee \neg\text{guilty}(c)) \\
&\wedge (\text{guilty}(a) \vee \neg\text{guilty}(d)) \\
&\wedge (\text{guilty}(a) \vee \neg\text{guilty}(e)) \\
&\wedge (\neg\text{guilty}(b) \vee \neg\text{guilty}(c)) \\
&\wedge (\neg\text{guilty}(b) \vee \neg\text{guilty}(d)) \\
&\wedge (\neg\text{guilty}(b) \vee \neg\text{guilty}(e)) \\
&\wedge (\neg\text{guilty}(c) \vee \neg\text{guilty}(d)) \\
&\wedge (\neg\text{guilty}(c) \vee \neg\text{guilty}(e)) \\
&\wedge (\neg\text{guilty}(d) \vee \neg\text{guilty}(e))
\end{aligned}$$

Resulting clauses:

$$\begin{aligned}
&\text{guilty}(a) \vee \text{guilty}(b) \vee \text{guilty}(c) \vee \text{guilty}(d) \vee \text{guilty}(e) \\
&\neg\text{guilty}(a) \vee \neg\text{guilty}(b) \\
&\neg\text{guilty}(a) \vee \neg\text{guilty}(c) \\
&\neg\text{guilty}(a) \vee \neg\text{guilty}(d) \\
&\neg\text{guilty}(a) \vee \neg\text{guilty}(e) \\
&\neg\text{guilty}(b) \vee \neg\text{guilty}(c) \\
&\neg\text{guilty}(b) \vee \neg\text{guilty}(d) \\
&\neg\text{guilty}(b) \vee \neg\text{guilty}(e) \\
&\neg\text{guilty}(c) \vee \neg\text{guilty}(d) \\
&\neg\text{guilty}(c) \vee \neg\text{guilty}(e) \\
&\neg\text{guilty}(d) \vee \neg\text{guilty}(e)
\end{aligned}$$

Bruno says: "The culprit is either Eduardo or Dorota".
Formula:

$$\text{guilty}(e) \vee (\text{guilty}(e) \rightarrow \text{guilty}(d))$$

Conversion to CNF:

$$\begin{aligned}
&\text{guilty}(e) \vee (\neg\text{guilty}(e) \vee \text{guilty}(d)) \rightarrow (\neg\text{guilty}(e) \vee \text{guilty}(d)) \\
&\wedge (\neg\text{guilty}(d) \vee \text{guilty}(d)) \rightarrow \text{guilty}(e) \\
&\wedge (\neg\text{guilty}(e) \vee \text{guilty}(e)) \rightarrow \text{guilty}(e) \\
&\wedge (\neg\text{guilty}(d) \vee \text{guilty}(e)) \rightarrow \text{guilty}(d) \\
&\wedge (\neg\text{guilty}(e) \vee \text{guilty}(d)) \rightarrow \text{guilty}(e) \\
&\wedge (\neg\text{guilty}(d) \vee \text{guilty}(d)) \rightarrow \text{guilty}(d) \\
&\wedge (\neg\text{guilty}(e) \vee \text{guilty}(e)) \rightarrow \text{guilty}(e) \\
&\wedge (\neg\text{guilty}(d) \vee \text{guilty}(e)) \rightarrow \text{guilty}(d)
\end{aligned}$$

Resulting clauses:

$$\begin{aligned}
&\text{guilty}(e) \vee \text{guilty}(d) \\
&\neg\text{guilty}(e) \vee \text{guilty}(d) \\
&\neg\text{guilty}(d) \vee \text{guilty}(d) \\
&\neg\text{guilty}(e) \vee \text{guilty}(e) \\
&\neg\text{guilty}(d) \vee \text{guilty}(e) \\
&\neg\text{guilty}(e) \vee \text{guilty}(d) \\
&\neg\text{guilty}(d) \vee \text{guilty}(d) \\
&\neg\text{guilty}(e) \vee \text{guilty}(e) \\
&\neg\text{guilty}(d) \vee \text{guilty}(e) \\
&\neg\text{guilty}(d) \vee \text{guilty}(d)
\end{aligned}$$

Eduardo says: "The guilty person is a girl".
Formula

$$\text{guilty}(e) \rightarrow (\text{female}(e) \wedge \text{guilty}(e))$$

Conversion to CNF:

$$\begin{aligned}
&\neg\text{guilty}(e) \vee (\text{female}(e) \wedge \text{guilty}(e)) \rightarrow (\neg\text{guilty}(e) \vee \text{female}(e)) \\
&\wedge (\neg\text{guilty}(e) \vee \text{guilty}(e)) \rightarrow \text{guilty}(e) \\
&\wedge (\neg\text{guilty}(e) \vee \text{female}(e)) \rightarrow \text{female}(e) \\
&\wedge (\neg\text{guilty}(e) \vee \text{guilty}(e)) \rightarrow \text{guilty}(e) \\
&\wedge (\neg\text{guilty}(e) \vee \text{female}(e)) \rightarrow \text{female}(e)
\end{aligned}$$

Resulting clauses:

$$\begin{aligned}
&\neg\text{guilty}(e) \vee \text{female}(e) \\
&\neg\text{guilty}(e) \vee \text{guilty}(e) \\
&\neg\text{guilty}(e) \vee \text{female}(e)
\end{aligned}$$

Dorota says: "If Bruno is guilty, then Carla is innocent".
Formula

$$\text{guilty}(b) \rightarrow (\text{guilty}(b) \rightarrow \neg\text{guilty}(c))$$

Conversion to CNF:

- Eliminate implications:

$$\neg\text{guilty}(b) \vee (\neg\text{guilty}(b) \vee \neg\text{guilty}(c))$$

- Expand equivalences:

$$\neg\text{guilty}(b) \vee (\neg\text{guilty}(b) \vee \neg\text{guilty}(c)) \wedge (\neg\text{guilty}(b) \vee \neg\text{guilty}(b)) \wedge (\neg\text{guilty}(b) \vee \neg\text{guilty}(c))$$

Resulting clauses:

$$\begin{aligned}
&\neg\text{guilty}(b) \vee \neg\text{guilty}(b) \vee \neg\text{guilty}(c) \\
&\neg\text{guilty}(b) \vee \neg\text{guilty}(b) \\
&\neg\text{guilty}(b) \vee \neg\text{guilty}(c)
\end{aligned}$$

4. Knowledge Base Clauses

After converting the formal representation of the knowledge base into CNF, we removed all duplicates and obtained the final unsharded list of clauses:

No	Knowledge Base Clause
1	$\text{guilty}(a) \vee \text{guilty}(b) \vee \text{guilty}(c) \vee \text{guilty}(d) \vee \text{guilty}(e)$
2	$\neg\text{guilty}(a) \vee \neg\text{guilty}(b)$
3	$\neg\text{guilty}(a) \vee \neg\text{guilty}(c)$
4	$\neg\text{guilty}(a) \vee \neg\text{guilty}(d)$
5	$\neg\text{guilty}(a) \vee \neg\text{guilty}(e)$
6	$\neg\text{guilty}(b) \vee \neg\text{guilty}(c)$
7	$\neg\text{guilty}(b) \vee \neg\text{guilty}(d)$
8	$\neg\text{guilty}(b) \vee \neg\text{guilty}(e)$
9	$\neg\text{guilty}(c) \vee \neg\text{guilty}(d)$
10	$\neg\text{guilty}(c) \vee \neg\text{guilty}(e)$
11	$\neg\text{guilty}(d) \vee \neg\text{guilty}(e)$
12	$\neg\text{guilty}(e) \vee \text{female}(e)$
13	$\neg\text{guilty}(e) \vee \text{guilty}(d)$
14	$\neg\text{guilty}(e) \vee \text{guilty}(e)$
15	$\neg\text{guilty}(d) \vee \text{guilty}(e)$
16	$\neg\text{guilty}(d) \vee \text{guilty}(d)$
17	$\neg\text{guilty}(b) \vee \neg\text{guilty}(c)$

5. Resolution with Set-of-Support Strategy

We **set-of-support** strategy[1] is a logical reference strategy in which resolution is applied only to clauses that involve a specially selected set of support, typically related to the goal being proved. This approach restricts the search space, reduces the reasoning process to relevant inferences, and helps reach a contradiction or the desired conclusion more efficiently.

Core idea:

- All clauses are divided into two sets:
  - **Knowledge Base (KB)** — trusted facts and rules
  - **Set of Support (SOS)** — clauses related to the negation of the goal

Resolution rule:

- Resolution is applied **only if at least one parent clause belongs to the SOS**

Why it works:

- Reduces the search space
- Prevents unnecessary resolutions
- Guides reasoning toward a contradiction or proof

Goal:

- Derive the **empty clause** () which confirms the conclusion

Result:

- More efficient and structured logical inference

5.1 Daniela

Now we are the hypothesis that **Daniela is innocent**, $\neg\text{guilty}(d)$. To do this, we apply the resolution method in order to confirm or refute this assumption:

No	Derived Clause	Resolution Step / Description
17	$\text{guilty}(d)$	Negated conclusion
18	$\text{guilty}(d) \vee \text{guilty}(e)$	Resolution of clauses 16 and 17
19	$\text{guilty}(e) \vee \text{guilty}(e)$	Resolution of clauses 18 and 17
20	$\neg\text{guilty}(d)$	Resolution of clauses 6 and 19
21	$\text{guilty}(e)$	Resolution of clauses 20 and 19
22	$\text{female}(e) \vee \text{guilty}(d)$	Resolution of clauses 12 and 19
23	$\text{female}(e) \vee \text{guilty}(e)$	Resolution of clauses 12 and 21
24	$\text{female}(e)$	Resolution of clauses 23 and 21
25	$\text{guilty}(d)$	Resolution of clauses 22 and 24
26	$\text{guilty}(e)$	Resolution of clauses 25 and 24
27	$\text{guilty}(d) \vee \text{guilty}(e)$	Resolution of clauses 13 and 26
28	$\text{guilty}(d)$	Resolution of clauses 27 and 26
29	$\text{guilty}(e)$	Resolution of clauses 27 and 28
30	$\neg\text{guilty}(c)$	Resolution of clauses 10 and 29
31	$\neg\text{guilty}(b)$	Resolution of clauses 8 and 29
32	$\neg\text{guilty}(b) \vee \neg\text{guilty}(c)$	Resolution of clauses 31 and 30
33	$\neg\text{guilty}(b)$	Resolution of clauses 32 and 30
34	$\text{guilty}(a) \vee \text{guilty}(b) \vee \text{guilty}(c) \vee \text{guilty}(d)$	Resolution of clauses 1 and 29
35	$\text{guilty}(a) \vee \text{guilty}(b) \vee \text{guilty}(c)$	Resolution of clauses 34 and 28
36	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 35 and 30
37	$\text{guilty}(a)$	Resolution of clauses 36 and 31
38	$\text{guilty}(a) \vee \text{guilty}(d)$	Resolution of clauses 37 and 4
39	$\text{guilty}(a) \vee \text{guilty}(e)$	Resolution of clauses 37 and 5
40	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 37 and 3
41	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 37 and 2
42	$\text{guilty}(a)$	Resolution of clauses 38 and 41
43	$\text{guilty}(a) \vee \text{guilty}(e)$	Resolution of clauses 42 and 5
44	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 42 and 3
45	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 42 and 2
46	$\text{guilty}(a)$	Resolution of clauses 43 and 44
47	$\text{guilty}(a) \vee \text{guilty}(d)$	Resolution of clauses 46 and 4
48	$\text{guilty}(a) \vee \text{guilty}(e)$	Resolution of clauses 46 and 5
49	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 46 and 3
50	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 46 and 2
51	$\text{guilty}(a)$	Resolution of clauses 47 and 50
52	$\text{guilty}(a) \vee \text{guilty}(e)$	Resolution of clauses 51 and 5
53	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 51 and 3
54	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 51 and 2
55	$\text{guilty}(a)$	Resolution of clauses 52 and 54
56	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 55 and 3
57	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 55 and 2
58	$\text{guilty}(a)$	Resolution of clauses 56 and 57
59	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 58 and 3
60	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 58 and 2
61	$\text{guilty}(a)$	Resolution of clauses 59 and 60
62	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 61 and 3
63	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 61 and 2
64	$\text{guilty}(a)$	Resolution of clauses 62 and 63
65	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 64 and 3
66	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 64 and 2
67	$\text{guilty}(a)$	Resolution of clauses 65 and 66
68	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 67 and 3
69	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 67 and 2
70	$\text{guilty}(a)$	Resolution of clauses 68 and 69
71	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 70 and 3
72	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 70 and 2
73	$\text{guilty}(a)$	Resolution of clauses 71 and 72
74	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 73 and 3
75	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 73 and 2
76	$\text{guilty}(a)$	Resolution of clauses 74 and 75
77	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 76 and 3
78	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 76 and 2
79	$\text{guilty}(a)$	Resolution of clauses 77 and 78
80	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 79 and 3
81	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 79 and 2
82	$\text{guilty}(a)$	Resolution of clauses 80 and 81
83	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 82 and 3
84	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 82 and 2
85	$\text{guilty}(a)$	Resolution of clauses 83 and 84
86	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 85 and 3
87	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 85 and 2
88	$\text{guilty}(a)$	Resolution of clauses 86 and 87
89	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 88 and 3
90	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 88 and 2
91	$\text{guilty}(a)$	Resolution of clauses 89 and 90
92	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 91 and 3
93	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 91 and 2
94	$\text{guilty}(a)$	Resolution of clauses 92 and 93
95	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 94 and 3
96	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 94 and 2
97	$\text{guilty}(a)$	Resolution of clauses 95 and 96
98	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 97 and 3
99	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 97 and 2
100	$\text{guilty}(a)$	Resolution of clauses 98 and 99
101	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 100 and 3
102	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 100 and 2
103	$\text{guilty}(a)$	Resolution of clauses 101 and 102
104	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 103 and 3
105	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 103 and 2
106	$\text{guilty}(a)$	Resolution of clauses 104 and 105
107	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 106 and 3
108	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 106 and 2
109	$\text{guilty}(a)$	Resolution of clauses 107 and 108
110	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 109 and 3
111	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 109 and 2
112	$\text{guilty}(a)$	Resolution of clauses 110 and 111
113	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 112 and 3
114	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 112 and 2
115	$\text{guilty}(a)$	Resolution of clauses 113 and 114
116	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 115 and 3
117	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 115 and 2
118	$\text{guilty}(a)$	Resolution of clauses 116 and 117
119	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 118 and 3
120	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 118 and 2
121	$\text{guilty}(a)$	Resolution of clauses 119 and 120
122	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 121 and 3
123	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 121 and 2
124	$\text{guilty}(a)$	Resolution of clauses 122 and 123
125	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 124 and 3
126	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 124 and 2
127	$\text{guilty}(a)$	Resolution of clauses 125 and 126
128	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 127 and 3
129	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 127 and 2
130	$\text{guilty}(a)$	Resolution of clauses 128 and 129
131	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 130 and 3
132	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 130 and 2
133	$\text{guilty}(a)$	Resolution of clauses 131 and 132
134	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 133 and 3
135	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 133 and 2
136	$\text{guilty}(a)$	Resolution of clauses 134 and 135
137	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 136 and 3
138	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 136 and 2
139	$\text{guilty}(a)$	Resolution of clauses 137 and 138
140	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 139 and 3
141	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 139 and 2
142	$\text{guilty}(a)$	Resolution of clauses 140 and 141
143	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 142 and 3
144	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 142 and 2
145	$\text{guilty}(a)$	Resolution of clauses 143 and 144
146	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 145 and 3
147	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 145 and 2
148	$\text{guilty}(a)$	Resolution of clauses 146 and 147
149	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 148 and 3
150	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 148 and 2
151	$\text{guilty}(a)$	Resolution of clauses 149 and 150
152	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 151 and 3
153	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 151 and 2
154	$\text{guilty}(a)$	Resolution of clauses 152 and 153
155	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 154 and 3
156	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 154 and 2
157	$\text{guilty}(a)$	Resolution of clauses 155 and 156
158	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 157 and 3
159	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 157 and 2
160	$\text{guilty}(a)$	Resolution of clauses 158 and 159
161	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 160 and 3
162	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 160 and 2
163	$\text{guilty}(a)$	Resolution of clauses 161 and 162
164	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 163 and 3
165	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 163 and 2
166	$\text{guilty}(a)$	Resolution of clauses 164 and 165
167	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 166 and 3
168	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 166 and 2
169	$\text{guilty}(a)$	Resolution of clauses 167 and 168
170	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 169 and 3
171	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 169 and 2
172	$\text{guilty}(a)$	Resolution of clauses 170 and 171
173	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 172 and 3
174	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 172 and 2
175	$\text{guilty}(a)$	Resolution of clauses 173 and 174
176	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 175 and 3
177	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 175 and 2
178	$\text{guilty}(a)$	Resolution of clauses 176 and 177
179	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 178 and 3
180	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 178 and 2
181	$\text{guilty}(a)$	Resolution of clauses 179 and 180
182	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 181 and 3
183	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 181 and 2
184	$\text{guilty}(a)$	Resolution of clauses 182 and 183
185	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 184 and 3
186	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 184 and 2
187	$\text{guilty}(a)$	Resolution of clauses 185 and 186
188	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 187 and 3
189	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 187 and 2
190	$\text{guilty}(a)$	Resolution of clauses 188 and 189
191	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 190 and 3
192	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 190 and 2
193	$\text{guilty}(a)$	Resolution of clauses 191 and 192
194	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 193 and 3
195	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 193 and 2
196	$\text{guilty}(a)$	Resolution of clauses 194 and 195
197	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 196 and 3
198	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 196 and 2
199	$\text{guilty}(a)$	Resolution of clauses 197 and 198
200	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 199 and 3
201	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 199 and 2
202	$\text{guilty}(a)$	Resolution of clauses 200 and 201
203	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 202 and 3
204	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 202 and 2
205	$\text{guilty}(a)$	Resolution of clauses 203 and 204
206	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 205 and 3
207	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 205 and 2
208	$\text{guilty}(a)$	Resolution of clauses 206 and 207
209	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 208 and 3
210	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 208 and 2
211	$\text{guilty}(a)$	Resolution of clauses 209 and 210
212	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 211 and 3
213	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 211 and 2
214	$\text{guilty}(a)$	Resolution of clauses 212 and 213
215	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 214 and 3
216	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 214 and 2
217	$\text{guilty}(a)$	Resolution of clauses 215 and 216
218	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 217 and 3
219	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 217 and 2
220	$\text{guilty}(a)$	Resolution of clauses 218 and 219
221	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 220 and 3
222	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 220 and 2
223	$\text{guilty}(a)$	Resolution of clauses 221 and 222
224	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 223 and 3
225	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 223 and 2
226	$\text{guilty}(a)$	Resolution of clauses 224 and 225
227	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 226 and 3
228	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 226 and 2
229	$\text{guilty}(a)$	Resolution of clauses 227 and 228
230	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 229 and 3
231	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 229 and 2
232	$\text{guilty}(a)$	Resolution of clauses 230 and 231
233	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 232 and 3
234	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 232 and 2
235	$\text{guilty}(a)$	Resolution of clauses 233 and 234
236	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 235 and 3
237	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 235 and 2
238	$\text{guilty}(a)$	Resolution of clauses 236 and 237
239	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 238 and 3
240	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 238 and 2
241	$\text{guilty}(a)$	Resolution of clauses 239 and 240
242	$\text{guilty}(a) \vee \text{guilty}(c)$	Resolution of clauses 241 and 3
243	$\text{guilty}(a) \vee \text{guilty}(b)$	Resolution of clauses 241 and 2
244	$\text{guilty}(a)$